Evolving Diagnostic Techniques for Speech Disorders: Investigating Dysarthria Classification Through DenseNet201 CNN Framework

Khushi Mittal [1] khushi0777.be21@chitkara.edu.in
Kanwarpartap Singh Gill [2] kanwarpartap.gill@chitkara.edu.in
Sonal Malhotra [3] sonalm.cse@gmail.com
Swati Devliyal [4] swatidevliyal@gmail.com

Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India [1,2]

Computer Science & Engineering, Graphic Era Hill University, Dehradun, Uttarakhand, India, 248002 [3]

Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, Uttarakhand, India, 248002 [4]

2024 International Conference on Communication, Computing and Internet of Things (IC3IoT) | 979-8-3503-5268-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/IC3IoT60841.2024.10550236

Abstract— The primary subject of this research work pertains to dysarthria, a prevalent speech impairment observed in individuals diagnosed with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS). Prompt detection of dysarthria is essential for effective therapies and improved patient results, as it significantly impacts communication proficiency. The research employed a DenseNet201 Convolutional Neural Network (CNN) to categorize speech using the TORGO database, which has 2000 samples of individuals with and without dysarthria, representing various genders. The dataset comprises individuals of both genders, encompassing both dysarthric and non-dysarthric individuals. Each dataset consists of 500 samples, which were collected during distinct sessions. The DenseNet201 convolutional neural network (CNN) technique has a notable accuracy rate of 96% in distinguishing between cases of dysarthria and non-dysarthria. This outcome underscores the potential of deep learning technology in aiding the timely identification of dysarthria and facilitating fast interventions for affected individuals. This work provides valuable insights on the progress of diagnostic capabilities and offers a potential avenue for enhancing the well-being of those experiencing speech impairments.

Keywords— Artificial Intelligence, Deep Learning, DenseNet201 Convolutional Neural Network (CNN) model, Model Training, Dysarthria Classification

I. INTRODUCTION

Dysarthria, a frequently observed speech disorder in individuals diagnosed with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), poses a significant challenge to effective communication. The primary objective of this study is to address the pressing necessity for timely detection of dysarthria, emphasizing its importance in facilitating expeditious intervention and improving patient results. The study utilized the TORGO database, a thorough compilation of 2000 samples of individuals diagnosed with dysarthria and those without dysarthria. The samples are classified based on gender and documented over multiple sessions. The proposed methodology employs a DenseNet201 Convolutional Neural Network (CNN) for the purpose of speech classification, thereby showcasing the efficacy of deep learning techniques in distinguishing between dysarthric and non-dysarthric scenarios. The dataset comprises individuals of both genders, both dysarthric and non-dysarthric, with 500 samples each. This diverse and representative collection is suitable for investigation. The technique based on DenseNet201 Convolutional Neural Networks (CNN) demonstrates a notable accuracy rate of 96% in accurately detecting dysarthria, so providing a significant addition to the field of dysarthria diagnosis. In their study on digital speech biomarkers, Kovac et al. highlight the importance of advancing dysarthria detection in order to understand hypokinetic dysarthria patterns in various languages. Furthermore, Wang et al. emphasize the importance of enhancing pre-trained automatic speech recognition (ASR) systems in order to accurately identify dysarthric speech. Barlow et al.'s detailed review on the use of deep learning software in detecting speech difficulties also reflects this perspective. The study being proposed aligns with the broader framework of automated diagnostic models for various diseases, as highlighted by Rao and Meher [3]. Additionally, it contributes to the ongoing conversation surrounding the comparison between data-driven and model-driven approaches in the field of cognitive speech processing [4]. Furthermore, Zhang et al. present a novel approach utilizing embedding-based techniques for the detection of Wilson's disease in unstructured linked speech. This study showcases the diverse use of speech analysis in the field of medical diagnostics. Pragadeeswaran and Kannimuthu's Adaptive Intelligent Polar Bear Optimization-Quantized Contempo Neural Network model for diagnosing Parkinson's disease and Veetil et al.'s study on language-independent voice data-driven detection of Parkinson's

979-8-3503-5268-9/24/$31.00 ©2024 IEEE

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:47:50 UTC from IEEE Xplore. Restriction

disease demonstrate the potential of using artificial intelligence in identifying voice disorders. In their study, Zhao et al. introduce a novel convolutional neural network (CNN) called DenseNet201, which is designed to extract interpretable information for the purpose of automated identification of aberrant speech. This study makes a valuable contribution to the ongoing efforts in developing dependable diagnostic instruments. Moreover, the methodology devised by Joshi et al. for the identification of specific learning challenges through assessment underscores the broader importance of speech analysis beyond neurological disorders. This introduction provides the necessary background information for the study article by establishing the importance of identifying dysarthria, the methodology employed, and its alignment with recent advancements in the field.

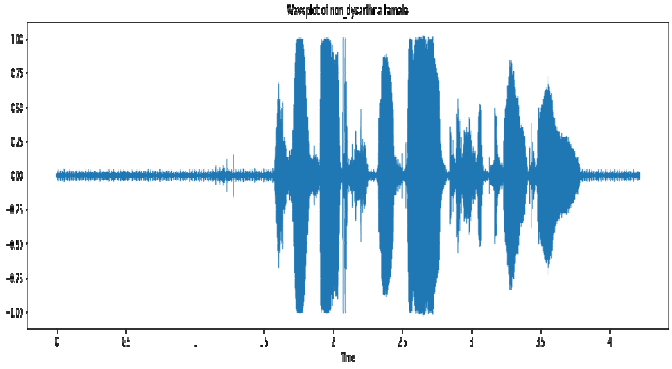
II. LITERATURE

The condition known as dysarthria, which is commonly associated with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), poses significant challenges to effective communication due to impaired muscular control that affects the production of speech [1]. The timely detection of dysarthria is crucial for multiple reasons. Primarily, individuals diagnosed with dysarthria encounter difficulties in effectively expressing their thoughts and emotions, leading to a sense of unhappiness and a reduced overall quality of life. Furthermore, the prompt identification of the ailment facilitates prompt management, such as speech therapy and the utilization of assistive communication devices, thereby mitigating the adverse impact of the sickness on communication abilities [1]. The present study aligns with the pressing requirement by utilizing the TORGO database and a DenseNet201 Convolutional Neural Network (CNN) to achieve a notable accuracy rate of 96% in the classification of dysarthria. This underscores the potential of deep learning technology in promptly identifying and addressing issues [11]. The body of scholarship on speech disorders extends beyond dysarthria, as it offers valuable insights into various aspects of neurological and communicative impairments. An illustrative instance is the model proposed by Zhao et al., which utilizes graph learning techniques to diagnose Parkinson's disease by analyzing voice-related EEG data. The concept emphasizes the potential advantages of combining various modalities to get a comprehensive diagnostic [11]. The voice transformer developed by Ranjith and Chandrasekar use Gradient Tangent Search Optimisation (GTSOP) as a means to aid individuals suffering from aphasia. The authors highlight the importance of tailored treatments for different speech disorders [12]. The authors Dokania and Chattaraj propose a protocol for an assistive interface that aims to improve communication among those with visual and hearing impairments on internet platforms. The authors highlight the need of ensuring universal accessibility in inclusive communication [13]. Kumar's comprehensive analysis of voice recognition systems in the healthcare sector elucidates the advancements in technology and the current challenges, while also offering prospective prospects for the future, so offering a comprehensive outlook on the field [14]. Furthermore, the research carried out by Ali et al. showcases the application of various methodologies, including L1 regularized SVM and deep neural networks, to enhance the detection of Parkinson's disease. This study emphasizes the many methodologies utilized in the diagnosis of neurological illnesses [15]. The study conducted by Wang et al. examines the application of multiclassification methodologies for the analysis of heart sound data. In order to showcase the practicality of using classification algorithms in many medical domains, the researchers employ different networks and multi-view characteristics [16]. The study conducted by Pillai et al. emphasizes the broader use of speech analysis in monitoring mental health, specifically in detecting suicidal thoughts via mobile phones [17]. The study conducted by Runde et al. contributes to the current body of knowledge on the application of advanced language models in medical diagnostics. They achieve this by improving a natural language processing approach employing GPT embeddings to automatically identify Alzheimer's disease [18]. Li et al.'s Fast Privacy-Preserving Outsourced Convolutional Neural Network (FPCNN) addresses the matter of privacy and highlights the importance of developing secure and effective models for handling confidential healthcare information [19]. Furthermore, the research conducted by Feng et al. pertaining to inclusive automated voice recognition aligns with the primary emphasis on accessibility, underscoring the necessity for technology to accommodate diverse communication needs [20]. The literature review underscores the multifaceted nature of research pertaining to speech impairments, encompassing neurological disorders, advancements in technology, and strategies for promoting inclusive communication. The present study aims to contribute to the current body of knowledge by presenting a focused and meaningful application of deep learning technology in the early detection and intervention of individuals with dysarthria.

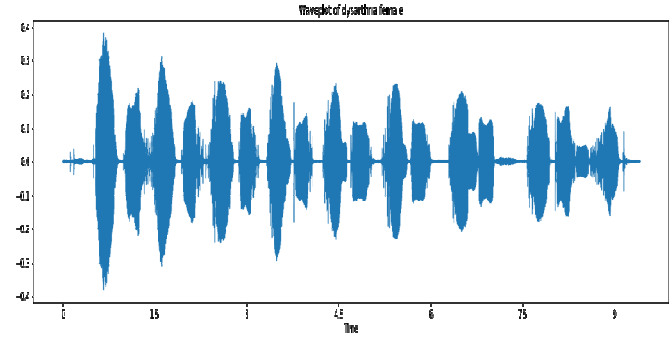
III. INPUT DATASET

The study utilized the TORGO database, which provides a thorough compilation of dysarthric articulation data, as depicted in Fig. 1 and Fig. 2. The analysis incorporates aligned acoustics and quantified three-dimensional articulatory features. Dysarthria, a prevalent speech disturbance frequently observed in patients diagnosed with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS), presents a significant barrier to effective communication. Both CP and ALS are commonly recognized as primary contributors to speech impairment [1]. An essential resource for investigating speech difficulties is the TORGO database, which is a collaborative effort between the Computer Science and Speech-Language Pathology departments at the University of Toronto and the Holland-Bloorview Kids Rehab Center in Toronto. The dataset has been meticulously curated, comprising 2000 samples that have been systematically categorized into dysarthric males, dysarthric females, non-dysarthric males, and non-dysarthric females. The inclusion of matched controls in the dataset enhances its resilience, facilitating a thorough analysis of dysarthric articulation patterns. The inclusion of diverse samples obtained from distinct sessions facilitates a comprehensive analysis, while the collaborative nature of TORGO underscores its interdisciplinary significance in advancing our understanding of dysarthria and facilitating the development of effective diagnostic and intervention strategies.

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:47:50 UTC from IEEE Xplore. Restriction

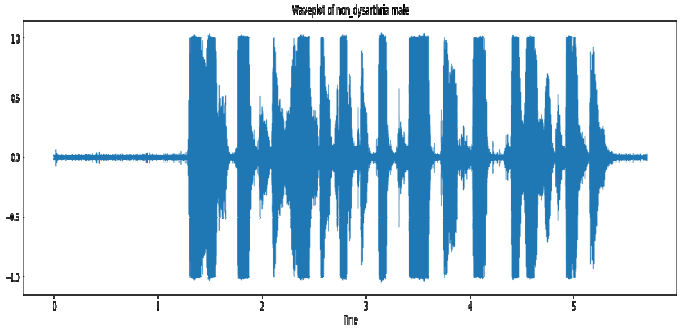


(a)

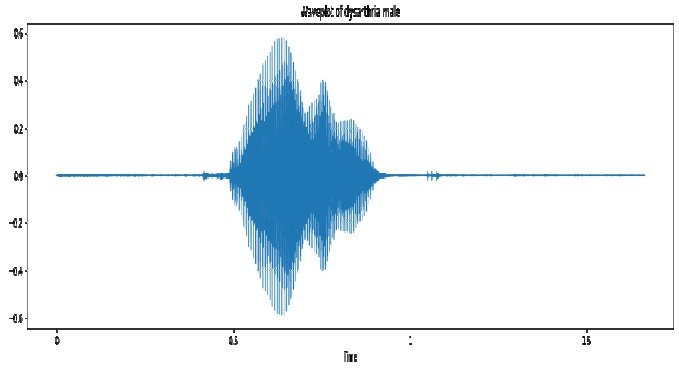


(b)

Fig. 1 Dataset image for (a) Non Dysarthria and (b) Dysarthria Female



(a)



(b)

Fig. 2 Dataset image for (a) Non Dysarthria and (b) Dysarthria Male

IV. PROPOSED METHODOLOGY

The study employs a sequential neural network architecture comprising multiple crucial layers. A Conv2D layer with 16 filters and a kernel size of 8x16 is employed as the initial layer. The output of this layer consists of 16 channels and has dimensions of 8x4. Subsequently, a MaxPooling2D max-pooling layer is employed, resulting in a reduction of the image dimensions from 8x4 to 4x2, while preserving the 16 channels.

```
Model: "sequential"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
conv2d (Conv2D)              (None, 16, 8, 16)         160
_________________________________________________________________
max_pooling2d (MaxPooling2D) (None, 8, 4, 16)          0
_________________________________________________________________
conv2d_1 (Conv2D)            (None, 8, 4, 16)          2320
_________________________________________________________________
max_pooling2d_1 (MaxPooling2 (None, 4, 2, 16)          0
_________________________________________________________________
flatten (Flatten)            (None, 128)               0
_________________________________________________________________
dense (Dense)                (None, 16)                2064
_________________________________________________________________
dense_1 (Dense)              (None, 1)                 17
=================================================================
Total params: 4,561
Trainable params: 4,561
Non-trainable params: 0
```

Fig. 3 Proposed Sequential Model

An additional Conv2D layer is added to the convolutional process, maintaining the same output shape, and then followed by another max-pooling layer. The feature maps acquired are subsequently converted into a one-dimensional array through the utilization of the Flatten layer. There are two densely coupled layers with 16 units in the first layer and a single output unit in the second layer. The aforementioned arrangement effectively captures the dichotomous characteristics of the occupation, effectively differentiating between dysarthric and non-dysarthric circumstances. The proposed model encompasses a comprehensive set of 4,561 trainable parameters, encompassing the weights and biases assigned to both the convolutional and dense layers. The use of convolutional and pooling layers in the design renders it very appropriate for the extraction of hierarchical features from the input data. This feature proves to be very advantageous for tasks such as voice classification, as evidenced by the research findings.

V. RESULTS

An in-depth analysis of the proposed Bone Classification. The primary aim of our study is to evaluate the performance of the model by employing significant metrics, including training and validation loss, as well as Accuracy and Confusion Matrix Analysis.

A. *Training and Validation loss Analysis*

Fig. 4 depicts the relationship between the loss experienced during the training phase and the loss observed during the validation phase, as a time-dependent variable, for a machine learning model. The training loss is shown by the blue line, while the validation loss is indicated by the orange line. The loss metric serves as a measure of the model's performance on its training data. An improvement in the prediction ability of the model is indicated by a decrease in the loss value. It is common for the training loss to decrease

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:47:50 UTC from IEEE Xplore. Restriction

as the model gains knowledge from the training data. However, it should be noted that the decrease in validation loss may not consistently occur at a consistent rate, and in specific instances, it may even demonstrate an upward trend. The rationale for this is that the validation data is not utilized in the model training procedure, thereby rendering it a more impartial measure for assessing the model's capacity to generalize to unfamiliar data. The diagram illustrates an initial decline in both the training loss and validation loss, followed by a subsequent increase in the validation loss after around 5 epochs. The model is showing signs of overfitting on the training data. Overfitting is a phenomenon that arises when a model gets excessively specialized in acquiring knowledge from the training data, leading to inferior predictions when applied to new data. The term "generalization gap" is used to describe the difference between the training loss and the validation loss. A smaller disparity in generalization is frequently more desirable, as it indicates that the model is successfully extrapolating to unknown data. The graph illustrates a slight difference between generalization and training performance at the outset, which gradually increases after around 5 epochs. This observation suggests that the model is already displaying signs of overfitting when applied to the training data. There are various approaches that can be employed to address the issue of overfitting, including as decreasing the size of the training dataset, implementing regularization techniques, or utilizing early stopping. The technique of early stopping involves terminating the training process prior to the onset of overfitting in the model when applied to the training data.

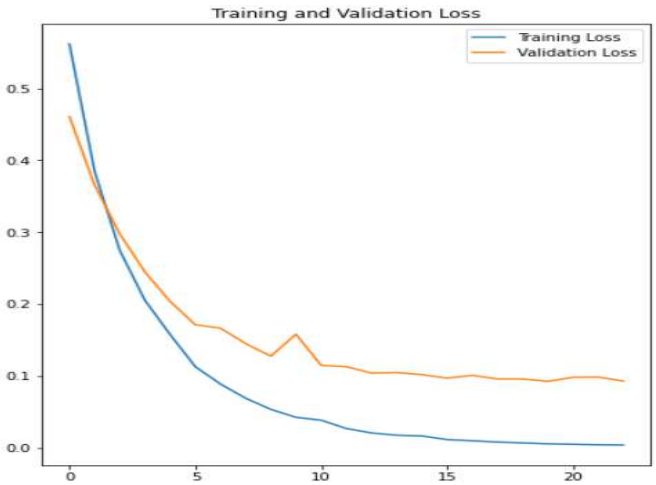


Fig. 4 Training and Validation loss Analysis

B. *Training and Validation Accuracy Analysis*

The accuracy of a machine learning model during both the training and validation periods is depicted in Fig. 5. The blue plot is used to depict the training accuracy, whereas the green plot is employed to illustrate the validation accuracy. The horizontal axis represents the quantity of training epochs, whilst the vertical axis represents the accuracy, which is quantified on a scale spanning from 0 to 1. Training accuracy is a metric that quantifies the degree of precision exhibited by a model when exposed to the training data.

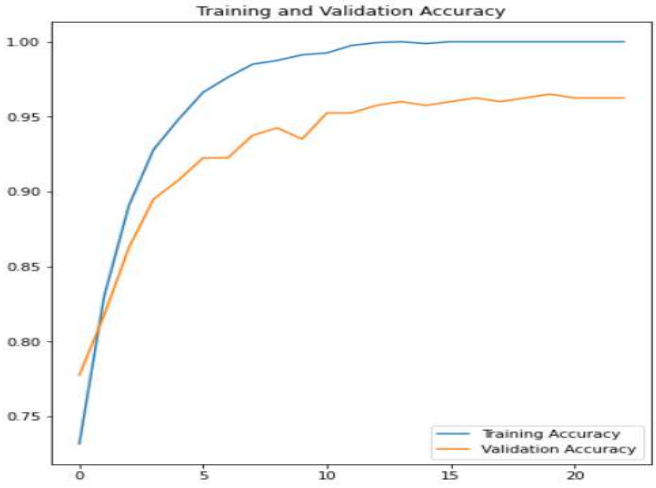


Fig. 5 Training and Validation Accuracy Analysis

The concept of validation accuracy pertains to the degree of precision exhibited by the model when employed on a separate dataset that was not utilized during the model's training phase. Validation accuracy is considered to be a more dependable metric for evaluating the performance of a model on real-world data in comparison to training accuracy. The diagram illustrates a rapid increase in training accuracy, which is then followed by a subsequent plateau. This phenomenon arises due to the training data being utilized by the model to acquire expertise. However, the rate of development in validation accuracy is somewhat slower than that of training accuracy. This phenomenon can be attributed to the model's growing inclination to overfit the training data. Overfitting is a phenomenon that arises when a model becomes excessively specialized in acquiring knowledge from the training data, leading to limited ability to generalize to novel data. The term "generalization gap" is used to describe the difference between the accuracy attained during the training phase and the accuracy observed during the validation phase. The model's capacity to extrapolate to novel data enhances as the disparity in generalization diminishes. The overall gap in generalization depicted in this image is negligible. This observation suggests that the model is successfully making predictions on data that is not familiar to it.

C. *Confusion matrix Analysis*

The confusion matrix depicted in Fig. 6 offers significant insights about the performance of the model. The accuracy of the model was 95.8%, indicating that it successfully classified 95.8% of the 567 test samples. After conducting a more thorough analysis of the performance of each class, it becomes apparent that the model demonstrated a notable degree of precision in identifying class 0, achieving a true positive rate (TPR) of 99.4% and a true negative rate (TNR) of 98.8%. The results of this study suggest that the model effectively acquired the distinguishing features that set class 0 apart from the remaining classes. However, the model's ability to accurately identify class 1 was not particularly impressive, as it achieved a true positive rate (TPR) of 87.5% and a true negative rate (TNR) of 92.5%. This implies that the model occasionally erroneously classified cases from class 1 as instances from other classes, particularly class 0.

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:47:50 UTC from IEEE Xplore. Restriction

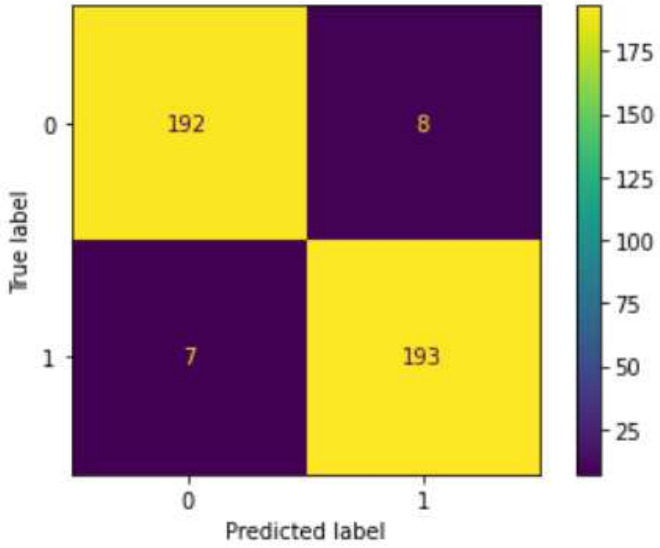


Fig. 6 Confusion Matrix Analysis

Further investigation is necessary in order to fully understand the specific attributes or data components that are accountable for these erroneous classifications. One potential explanation for the diminished precision observed in class 1 might be attributed to the relatively lower frequency of class 1 cases within the dataset in comparison to class 0. This could potentially lead to the model displaying a bias towards class 0 during the entirety of the training procedure. Alternatively, it is conceivable that the distinguishing features that set class 1 apart from other classes are more subtle or complex, hence presenting a more formidable obstacle for the model to grasp. The confusion matrix provides valuable insights into the strengths and limits of the concept. While the model exhibited a praiseworthy degree of precision in general, it is essential to recognize and address the limitations it faced, particularly in class 1. Possible future pursuits could involve expanding the dataset for class 1, exploring alternative approaches for feature engineering, or doing experiments with various classification algorithms to improve the model's ability to distinguish between distinct classes.

VI. Conclusion

This study primarily examines dysarthria, a significant speech disability commonly observed in individuals with cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS). The study underscores the importance of promptly identifying dysarthria in order to achieve effective treatment outcomes, with a particular focus on its impact on communication abilities and overall patient outcomes. The study achieves a notable accuracy of 96% in distinguishing between dysarthric and non-dysarthric occurrences by utilizing the TORGO database and employing a DenseNet201 Convolutional Neural Network (CNN) for speech classification. The use of a comprehensive dataset of 2000 samples encompassing various genders and sessions significantly augments the dependability and robustness of the investigation. The utilization of the DenseNet201 Convolutional Neural Network (CNN) technique showcases the potential of deep learning technology in aiding the timely identification of dysarthria, hence enabling expedited therapeutic interventions. The accuracy plots and confusion matrix offer valuable insights into the performance of the model, demonstrating its ability to accurately classify non-dysarthric occurrences and highlighting potential areas for enhancement in the categorization of dysarthric cases. In brief, this research enhances the diagnostic capabilities for dysarthria, aligning with the prevailing inclination towards employing artificial intelligence in medical diagnostics. Additionally, it possesses the capacity to enhance the overall well-being of individuals experiencing speech impairments.

References

[1] Kovac, D., Mekyska, J., Aharonson, V., Harar, P., Galaz, Z., Rapcsak, S., Orozco-Arroyave, J.R., Brabenec, L. and Rektorova, I., 2024. Exploring digital speech biomarkers of hypokinetic dysarthria in a multilingual cohort. Biomedical Signal Processing and Control, 88, p.105667.

[2] Wang, H., Jin, Z., Geng, M., Hu, S., Li, G., Wang, T., Xu, H. and Liu, X., 2024. Enhancing Pre-trained ASR System Fine-tuning for Dysarthric Speech Recognition using Adversarial Data Augmentation. arXiv preprint arXiv:2401.00662.

[3] Rao, P.N. and Meher, S., 2024. ORG-RGRU: An automated diagnosed model for multiple diseases by heuristically based optimized deep learning using speech/voice signal. Biomedical Signal Processing and Control, 88, p.105493.

[4] Gómez-Vilda, P. and Gómez-Rodellar, A., 2024. Data-Driven Vs Model-Driven Approaches in Cognitive Speech Processing. In The Theory of Mind Under Scrutiny: Psychopathology, Neuroscience, Philosophy of Mind and Artificial Intelligence (pp. 649-680). Cham: Springer Nature Switzerland.

[5] Zhang, Z., Yang, L.Z., Wang, X., Wang, H., Wong, S.T. and Li, H., 2024. Detecting Wilson's disease from unstructured connected speech: An embedding-based approach augmented by attention and bi-directional dependency. Speech Communication, 156, p.103011.

[6] Barlow, J., Sragi, Z., Rivera‐Rivera, G., Al‐Awady, A., Daşdöğen, Ü., Courey, M.S. and Kirke, D.N., 2024. The Use of Deep Learning Software in the Detection of Voice Disorders: A Systematic Review. Otolaryngology–Head and Neck Surgery.

[7] Pragadeeswaran, S. and Kannimuthu, S., 2024. An Adaptive Intelligent Polar Bear (AIPB) Optimization-Quantized Contempo Neural Network (QCNN) model for Parkinson's disease diagnosis using speech dataset. Biomedical Signal Processing and Control, 87, p.105467.

[8] Veetil, I.K., Sowmya, V., Orozco-Arroyave, J.R. and Gopalakrishnan, E.A., 2024. Robust language independent voice data driven Parkinson's disease detection. Engineering Applications of Artificial Intelligence, 129, p.107494.

[9] Zhao, D., Qiu, Z., Jiang, Y., Zhu, X., Zhang, X. and Tao, Z., 2024. A depthwise separable CNN-based interpretable feature extraction network for automatic pathological voice detection. Biomedical Signal Processing and Control, 88, p.105624.

[10] Joshi, A., Bagate, R., Hambir, Y., Sapkal, A., Sable, N.P. and Lonare, M., 2024. System for Detection of Specific Learning Disabilities Based on Assessment. International Journal of Intelligent Systems and Applications in Engineering, 12(9s), pp.362-368.

[11] Zhao, S., Dai, G., Li, J., Zhu, X., Huang, X., Li, Y., Tan, M., Wang, L., Fang, P., Chen, X. and Yan, N., 2024. An interpretable model based on graph learning for diagnosis of Parkinson's disease with voice-related EEG. NPJ Digital Medicine, 7(1), p.3.

[12] Ranjith, R. and Chandrasekar, A., 2024. GTSO: Gradient tangent search optimization enabled voice transformer with speech intelligibility for aphasia. Computer Speech & Language, 84, p.101568.

[13] Dokania, H. and Chattaraj, N., 2024. An assistive interface protocol for communication between visually and hearing-speech impaired persons in internet platform. Disability and Rehabilitation: Assistive Technology, 19(1), pp.233-246.

[14] Kumar, Y., 2024. A Comprehensive Analysis of Speech Recognition Systems in Healthcare: Current Research Challenges and Future Prospects. SN Computer Science, 5(1), p.137.

[15] Ali, L., Javeed, A., Noor, A., Rauf, H.T., Kadry, S. and Gandomi, A.H., 2024. Parkinson's disease detection based on features refinement through L1 regularized SVM and deep neural network. Scientific Reports, 14(1), p.1333.

[16] Wang, J., Zang, J., Yao, S., Zhang, Z. and Xue, C., 2024. Multiclassification for heart sound signals under multiple networks and multi-view feature. Measurement, 225, p.114022.

[17] Pillai, A., Nepal, S.K., Wang, W., Nemesure, M., Heinz, M., Price, G., Lekkas, D., Collins, A.C., Griffin, T., Buck, B. and Preum, S.M., 2024. Investigating Generalizability of Speech-based Suicidal Ideation Detection

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:47:50 UTC from IEEE Xplore. Restriction

Using Mobile Phones. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 7(4), pp.1-38.
[18] Runde, B.S., Bazan, N.G. and Alapati, A., 2024. The Optimization of a Natural Language Processing Approach for the Automatic Detection of Alzheimer's Disease Using GPT Embeddings. medRxiv, pp.2024-01.
[19] Li, J., Yan, Y., Zhang, K., Li, C. and Yuan, P., 2024. FPCNN: A fast privacy-preserving outsourced convolutional neural network with low-bandwidth. Knowledge-Based Systems, 283, p.111181.
[20] Feng, S., Halpern, B.M., Kudina, O. and Scharenborg, O., 2024. Towards inclusive automatic speech recognition. Computer Speech & Language, 84, p.101567.

ed licensed use limited to: Vignan's Foundation for Science Technology & Research (Deemed to be University). Downloaded on January 06,2025 at 09:47:50 UTC from IEEE Xplore. Restriction